

Fig. 3: Front view of Nokia LCD with pin details of the connector

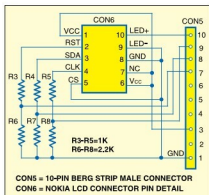


Fig. 4: Circuit for Nokia LCD connector extension

ant JTAG test interface (also used for accessing the on-chip debug system and programming) and six software-selectable power-saving modes.

PCF8833 LCD driver

The PCF8833 is a single-chip, low-power CMOS LCD controller driver, designed to drive colour super-twisted nematic displays of 132 rows and 132 RGB columns. All the necessary functions for the display are provided in a single chip, including the display RAM which has a capacity of 209 kbits. The PCF8833 uses multiple-row addressing technique in order to achieve the best optimal performance at the lowest power consumption. It offers two types of microcontroller interfaces: 8080 sys-

R	R	R	R	R	G	G	G	G	G
B	B	B	B	B	R	R	R	R	R
G	G	G	G	G	B	B	B	B	B

Fig. 5: Standard 12-bit image



Fig. 6: GUI created using VB.net for image transfer from the PC

tem interface and 3-line serial interface.

The PCF8833 communicates with the host using an 8-bit parallel interface or 3-line serial interface. Here a 3-line serial interface is implemented for communication between the microcontroller and the PCF8833 chip. The three lines are chip-select (CS) or enable pin, serial clock (SCLK) and serial data (SD).

Processing the instructions and data sent to the interface does not require the display clock. The display clock and interface clock are independent of each other. The display clock is derived from the built-in oscillator. The serial data pin of PCF8833 is connected to the SDA pin of the microcontroller.

Circuit description

Fig. 2 shows the circuit for interfacing the Nokia colour LCD with Atmega2560 microcontroller. The circuit is powered from a standard 9V DC source. The 9V DC is converted into 5V DC using a 7805 regulator (IC1). The glowing of LED1 shows the presence of power in the circuit. The regulated 5V supply powers the circuit including the Atmega2560, MAX232, LCD connector (CON2) and ISP6 connector (CON4). CON3 is a 9-pin, D-type COM port connector used to interface with the PC for picture file transfer to the LCD through the microcontroller (IC2). ISP6 connector is used for programming the microcontroller using STK500 board.

Whenever the program doesn't function properly, you can reset the microcontroller by momentarily

pressing reset switch S1.

Pin details of Nokia's 10-pin LCD connector are shown in Fig. 3. A manual implementation of the SPI with the microcontroller is done through SDA, CLK and CS pins on the LCD connector. First, the clock pin is cleared, then the data pin is set or cleared depending on the data bit and the clock pin set. SDA and CLK pins of the LCD connector are connected to pins 77 (PA1) and 76 (PA2) of the microcontroller, respectively.

The 8-bit data is transferred through the SPI in eight clock cycles. CS pin of the LCD chip is permanently made low. Backlight LED pin for the LCD may be connected to a PWM pin to have a dimming effect depending on the duty cycle of PWM.

The LED and Reset pins of the LCD are connected to pins 75 (PA3) and 78 (PA0) of the microcontroller, respectively. When PA0 is made low, it resets the LCD. All the lines PA0 to PA3 is made an output pin by setting the data direction register (DDRA) bits high.

This connector is too small to handle and solder using a normal soldering iron. You can extend the connections to a standard-size berg strip male connector as shown in Fig. 4. The LCD connector extension may be developed with resistors R3 through R5 (1 kilo-ohm each) and R6 through R8 (2.2 kilo-ohms each) forming the voltage divider circuits to obtain 3.3V signals (acceptable by the colour LCD) from 5.5V TTL signals.

Note that this connector extension is made for use with a colour LCD as well as standard character LCD. The LCD receives 3.3V supply through preset VR1. VR1 is also used to adjust the brightness of the colour LCD. Contrast is adjusted through software program.

Make a PCB layout and glue the board on the back side of the LCD module. Now you can easily fit the LCD onto the 10-pin berg strip socket